

#1: What is React:

React lets you **break the UI** into small reusable pieces called **components**.

- A component has it's own JS , CSS and HTML ;
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#2: What is JSX:

JSX(JavaScript Extension Syntax) = It lets us write **HTML directly inside JS**

- JSX is not understood by **browsers** directly.
 - so **Babel** transpile JSX into JS.
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#3: Step to setup the local Environment for react:



Prerequisites

Make sure you have installed:

- Node.js (LTS version)
- npm (comes with Node.js)

Check versions:

```
node -v  
npm -v
```



Step 1: Create React Project using Vite

Run this command:

```
npm create vite@latest
```



Step 2: Project Setup Questions

You will be asked:

- Project name → (your choice)
- Framework → **React**
- Variant → **JavaScript** (or TypeScript if you want)

Example:

- ✓ Project **name:** my-app
- ✓ **Framework:** React
- ✓ **Variant:** JavaScript



Step 3: Move into Project Folder

```
cd my-app
```



Step 4: Install Dependencies

```
npm install
```



Step 5: Run Development Server

```
npm run dev
```

Step 6: Open in Browser

After running, you will see something like:

<http://localhost:5173/>

Open it in your browser.

#4: Understanding our React App:

Important files:

1. `src/App.css`

CSS specifically for `App.jsx` (for **component-specific-styling**)

2. `src/App.jsx`

Main house where all rooms (**components**) live

3. `src/index.css`

Applies styles to **entire application** (for `index.html`)

4. `src/main.jsx` - (***DON'T TOUCH IT***)

Mount the `App.jsx` component into the `index.html` page

5. `index.html` - (***DON'T TOUCH IT***)

Root HTML file: Single HTML page of your entire React app, React injects everything inside this `root` .

#6: How to create and render Component:

Component is a reusable & independent piece of code.

Creating a component

```
function Title() {  
  return (  
    <h1>Hello World! </h1>  
  );  
}
```

Rendering a Component (Two Ways)

1. `<Title></Title>` or
2. `<Title />`

#7: Import and Export (Component / Files):

1. (Default) Export and Import:

Default Export:

```
export default Title;
```

A file can have **only one default export**

Import:

```
import Title from "./components/Title";
```

No `{}` needed, and you can rename it

2. (Named) Export and import:

Named Export:

```
export {Title};
```

We can export **multiple things** from a file.

Import:

```
import {Title} from "./components/Title.jsx";
```

We must use **same name** inside curly braces `{}`

#8: Writing Markup in JSX:

RULE 1:

Return a sing root element, Use Fragment (`<> </>`) to avoid extra div

RULE 2:

Always close all the tags properly.

RULE 3:

camelCase most of the things - `class` → `className` .

#9: React Fragment:

Fragments let you group a list of children without adding extra nodes to the DOM. If you don't want extra `<div>` in HTML:

```
return (  
  <>  
    <h1>Hello</h1>  
    <p>World</p>  
  </>  
);
```

#10: JSX with Curly Braces:

In React, JSX allows you to write **JavaScript** inside **HTML-like code** using **curly braces** `{}` .

- Curly braces let you **embed JavaScript expressions** inside **JSX**.

Basic Example

```
const name = "Pawan";  
  
function App() {  
  return <h1>Hello {name}</h1>;  
}
```

👉 Output: Hello Pawan